

Aptamer-Based Cancer Cell Analysis and Treatment.

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Aptamers are a class of single-stranded DNA or RNA oligonucleotides that can exclusively bind to various targets with high affinity and selectivity. Regarded as "chemical antibodies", aptamers possess several intrinsic advantages, including easy synthesis, convenient modification, high programmability, and good biocompatibility. In recent decades, many studies have demonstrated the superiority of aptamers as molecular tools for various biological applications, particularly in the area of cancer theranostics. In this review, we focus on recent progress in developing aptamer-based strategies for the precise analysis and treatment of cancer cells.

Aptamers as a novel tool for diagnostics and therapy.

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Aptamers are short single-stranded DNA or RNA oligonucleotides that are capable of binding small molecules, proteins, or nucleotides with high specificity. They show a stable conformation and high binding affinity for their target molecules. There are numerous applications for aptamers in biotechnology, molecular diagnostics and targeted therapy of diseases. Their production is cheap, and they generally display lower immunogenicity than monoclonal antibodies. In the present review, we give an introduction to the preparation of aptamers and provide examples for their use in biotechnology, diagnostics and therapy of diseases.

Aptamer Oligonucleotides as Potential Therapeutics in Hematologic Diseases.

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Aptamers are single-stranded DNA or RNA oligonucleotides generated by a novel in vitro selection technique termed Systematic evolution of ligands by exponential enrichment (SELEX). During the past two decades, various aptamer drugs have been developed and many of them have entered into clinical trials. In the present review, we focus on aptamers as potential therapeutics for hematological diseases, including anemia of chronic inflammation (ACI) and anemia of chronic disease (ACD), hemophilia, thrombotic thrombocytopenic purpura (TTP) or VWD type-2B, and sickle cell disease (SCD), in particular, those that have entered into clinical trials.

Aptamer applications for targeted cancer therapy.

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Aptamers are single-stranded DNA or RNA oligonucleotides that assume specific 3D structures and bind to target molecules with high affinity. The unique specificity of aptamers has made them attractive agents for targeted cancer therapy. Aptamers have been developed against a variety of cancer targets, including extracellular ligands and cell surface proteins. In addition, aptamers have been incorporated into novel constructs involving siRNAs, chemotherapeutic agents, cell toxins and nanoparticles, in which

they function as delivery agents for therapeutic cargo. In this article, we review recent developments in the use of aptamers for targeted cancer therapy, particularly focusing on novel applications of aptamers targeting the cell surface.

Aptamers: Biomedical Interest and Applications.

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Aptamers are short DNA or RNA oligonucleotides specialized in the specific and efficient binding to a target molecule. They are obtained by in vitro selection or evolution processes. It was in 1990 that two independent research groups described the bases of a new in vitro technology for the identification of RNA molecules able to specifically bind to a target [1,2]. Tuerk and Gold established the principals of the in vitro selection process that was named SELEX (Systematic Evolution of Ligands by Exponential enrichment), which is based on iterative cycles of binding, partitioning, and amplification of oligonucleotides from a pool of variant sequences [2]. Ellington and Szostak coined the term aptamer to define the selected molecules by the application of this method [1]. To date, numerous reports have described the isolation of aptamers directed against a great variety of targets covering a wide diversity of molecules varying in nature, size, and complexity ranging from ions to whole cells, including small molecules (e.g., aminoacids, nucleotides, antibiotics), peptides, proteins, nucleic acids, and viruses, among others (for example, see [3-6]). Modifications and optimization of the SELEX procedure aimed to get newly modified aptamers has also attracted much interest (examples can be found in [7,8]). These advances along with the parallel progresses in the nucleic acids chemistry and cellular delivery fields have allowed for the rise of a new hope in developing aptamers as efficient molecular tools for diagnostics and therapeutics (for recent comprehensive reviews, see [9-11]).

Recent Advances in Aptamer Sensors.

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Recently, aptamers have attracted attention in the biosensing field as signal recognition elements because of their high binding affinity toward specific targets such as proteins, cells, small molecules, and even metal ions, antibodies for which are difficult to obtain. Aptamers are single oligonucleotides generated by in vitro selection mechanisms via the systematic evolution of ligand exponential enrichment (SELEX) process. In addition to their high binding affinity, aptamers can be easily functionalized and engineered, providing several signaling modes such as colorimetric, fluorometric, and electrochemical, in what are known as aptasensors. In this review, recent advances in aptasensors as powerful biosensor probes that could be used in different fields, including environmental monitoring, clinical diagnosis, and drug monitoring, are described. Advances in aptamer-based colorimetric, fluorometric, and electrochemical aptasensing with their advantages and disadvantages are summarized and critically discussed. Additionally, future prospects are pointed out to facilitate the development of aptasensor technology for different targets.

In vitro RNA SELEX for the generation of chemically-optimized therapeutic RNA drugs.

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Aptamers are single-stranded DNA or RNA oligonucleotides that can bind with exquisitely high affinity and specificity to target molecules and are thus often referred to as 'nucleic acid' antibodies. Oligonucleotide aptamers are derived through a process of directed chemical evolution called SELEX (Systematic Evolution of Ligands by Exponential enrichment). This chemical equivalent of Darwinian evolution was first described in 1990 by Tuerk & Gold and Ellington & Szostak and has since yielded aptamers for a wide-range of applications, including biosensor technologies, in vitro diagnostics, biomarker discovery, and therapeutics. Since the inception of the original SELEX method, numerous modifications to the protocol have been described to fit the choice of target, specific conditions or applications. Technologies such as high-throughput sequencing methods and microfluidics have also been adapted for SELEX. In this chapter, we outline key steps in the SELEX process for enabling the rapid identification of RNA aptamers for in vivo applications. Specifically, we provide a detailed protocol for the selection of chemically-optimized RNA aptamers using the original in vitro SELEX methodology. In addition, methods for performing next-generation sequencing of the RNAs from each round of selection, based on Illumina sequencing technology, are discussed.

Aptamers: a promising tool for cancer imaging, diagnosis, and therapy.

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Aptamers are a group of molecules, which can specifically bind, track, and inhibit target molecules, comprising DNA aptamers, RNA aptamers, and peptide aptamers. So far, there are much progress about developing novel aptamers and their expansile applications. This prospect systematically introduces the composition and technological evolution of aptamers, and then focuses on the application of aptamers in cancer diagnosis, imaging, and therapy. Following this, we discuss the potential to harness aptamers in discovering the biomarker of stem cells, which is favorable for us to study the normal developmental or abnormal pathological process of tissue and to deliver drugs into target cells or tissues in the future.

Designing of peptide aptamer targeting the receptor-binding domain of spike protein of SARS-CoV-2: an in silico study.

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Short synthetic peptide molecules which bind to a specific target protein with a high affinity to exert its function are known as peptide aptamers. The high specificity of aptamers with small-molecule targets (metal ions, dyes and theophylline; ATP) is within 1 pM and 1 μ M range, whereas with the proteins (thrombin, CD4 and antibodies) it is in the nanomolar range (which is equivalent to monoclonal antibodies). The recently identified coronavirus (SARS-CoV-2) genome encodes for various proteins, such as envelope, membrane, nucleocapsid, and spike protein. Among these, the protein necessary for the virus to enter inside the host cell is spike protein. The work focuses on designing peptide aptamer targeting the spike receptor-binding domain of SARS-CoV-2. The peptide aptamer has been designed by using bacterial Thioredoxin A as the scaffold protein and an 18-residue-long peptide. The tertiary structure of the peptide aptamer is modeled and docked to spike receptor-binding domain of SARS CoV2. Molecular dynamic simulation has been done to check the stability of the aptamer and receptor-binding domain complex. It was observed that the aptamer binds to spike receptor-binding domain of

SARS-CoV-2 in a similar pattern as that of ACE2. The aptamer-receptor-binding domain complex was found to be stable in a 100 ns molecular dynamic simulation. The aptamer is also predicted to be non-antigenic, non-allergenic, non-hemolytic, non-inflammatory, water-soluble with high affinity toward ACE2 than serum albumin. Thus, peptide aptamer can be a novel approach for the therapeutic treatment for SARS-CoV-2.

The use of aptamers in large arrays for molecular diagnostics.

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Aptamers are single-stranded oligonucleotides derived from an in vitro evolution protocol called systematic evolution of ligands by exponential enrichment (SELEX). They bind tightly and specifically to target molecules; most aptamers to proteins bind with Kds (equilibrium dissociation constant) in the range of 1 pM to 1 nM. The SELEX protocol has been automated; therefore, hundreds to thousands of aptamers can be made in an economically feasible fashion. Blood and urine can be analyzed on chips that capture and quantitate proteins. SELEX has been adapted to the use of 5-bromo (5-Br) and 5-iodo (5-I) deoxyuridine residues. These halogenated bases can be specifically cross-linked to proteins. Selection pressure during in vitro evolution can be applied for both binding specificity and specific photo-cross-linkability. These are sufficiently independent parameters to allow one reagent, a photo-cross-linkable aptamer, to substitute for two reagents, the capture antibody and the detection antibody, in a typical sandwich array. After a cycle of binding, washing, cross-linking, and detergent washing, proteins will be specifically and covalently linked to their cognate aptamers. Because no other proteins are present on the chips, protein-specific stain will now show a meaningful array of pixels on the chip. Learning algorithms and retrospective studies should lead to a robust, simple, diagnostic chip.
